Results ledger: what was tested and what came out (exploratory, 3 mice, n = unflagged sessions unless stated)

what was tested	effect [95% CI] unit	sessions / mice	verdict
Fig. 2 Coupling of the axon mean to behaviour			
peak r axon mean vs face motion (±10 s)	+0.48 [0.28, 0.67] r	13 / 3	SUPPORTED
peak r axon mean vs whisking (±10 s)	+0.48 [0.26, 0.64] r	13 / 3	SUPPORTED
peak r axon mean vs pupil area (±10 s)	+0.44 [0.23, 0.62] r	13 / 3	SUPPORTED
peak r axon mean vs eyelid (±10 s)	+0.35 [0.20, 0.50] r	13 / 3	AMBIGUOUS
peak r axon mean vs ball speed (±10 s)	+0.56 [0.47, 0.65] r	4 / 1	AMBIGUOUS
net sign of single-ROI coupling to face motion: signal-like ROIs	+0.92 [0.78, 1.00] net sign	13 / 3	SUPPORTED
net sign of single-ROI coupling to whisking: signal-like ROIs	+0.90 [0.78, 0.98] net sign	13 / 3	SUPPORTED
net sign of single-ROI coupling to pupil area: signal-like ROIs	+0.94 [0.86, 1.00] net sign	13 / 3	SUPPORTED
held-out R2 of the axon mean from behaviour (ridge, lags -1..+3 s)	+0.34 [0.14, 0.52] R2	13 / 3	AMBIGUOUS
Fig. 3 Arousal events and state tuning			
axon response (0..5 s minus -6..-1 s, session z) to face-motion bout onset	+0.81 [0.26, 1.31] z	13 / 3	SUPPORTED
axon response (0..5 s minus -6..-1 s, session z) to whisking bout onset	+0.66 [0.08, 1.30] z	13 / 3	SUPPORTED
axon response (0..5 s minus -6..-1 s, session z) to pupil dilation onset	+0.49 [0.03, 0.97] z	10 / 3	AMBIGUOUS
axon response (0..5 s minus -6..-1 s, session z) to locomotion onset (ball)	+1.26 [1.07, 1.50] z	3 / 1	AMBIGUOUS
axon response (0..2 s minus -3..-1 s, session z) to saccade-like eye movement	+0.12 [0.04, 0.20] z	13 / 3	AMBIGUOUS
active − quiet (face-motion terciles) (dF/F)	+0.12 [0.03, 0.24] dF/F	13 / 3	SUPPORTED
running − still (ball) (dF/F)	+0.22 [0.09, 0.40] dF/F	4 / 1	AMBIGUOUS
Fig. 4 ROI structure			
PC1 share of variance among signal-like ROIs (band-passed 1-30 s)	+0.59 [0.45, 0.68] frac.	13 / 3	SUPPORTED
|r| of PC2 of the signal-like ROIs with face motion	+0.24 [0.10, 0.39] |r|	13 / 3	AMBIGUOUS
Spearman rho between pairwise ROI correlation and ROI distance	-0.16 [-0.40, 0.17] rho	12 / 3	AMBIGUOUS
Fig. 5 Visual stimuli			
axon response to full-field / local flash (raw)	+0.056 [0.020, 0.124] dF/F	5 / 3	AMBIGUOUS
axon response to full-field / local flash (movement removed)	+0.035 [0.007, 0.073] dF/F	5 / 3	AMBIGUOUS
axon response to moving bar (raw)	-0.003 [-0.020, 0.024] dF/F	5 / 3	NOT SEEN
axon response to moving bar (movement removed)	-0.002 [-0.024, 0.010] dF/F	5 / 3	NOT SEEN
axon response to expanding loom (raw)	+0.022 [-0.048, 0.135] dF/F	5 / 3	AMBIGUOUS
axon response to expanding loom (movement removed)	+0.004 [-0.037, 0.069] dF/F	5 / 3	NOT SEEN
axon response to receding loom (raw)	+0.008 [-0.003, 0.027] dF/F	4 / 3	NOT SEEN
axon response to receding loom (movement removed)	+0.010 [0.001, 0.028] dF/F	4 / 3	AMBIGUOUS
axon response to square (task / passive) (raw)	+0.006 [-0.004, 0.014] dF/F	7 / 2	NOT SEEN
axon response to square (task / passive) (movement removed)	+0.000 [-0.017, 0.011] dF/F	7 / 2	NOT SEEN
expanding minus receding loom response (raw)	-0.011 [-0.048, 0.056] dF/F	4 / 3	NOT SEEN
expanding minus receding loom response (movement removed)	-0.002 [-0.042, 0.065] dF/F	4 / 3	NOT SEEN
trial-wise expanding-loom axon response vs evoked pupil change (within-session Spearman)	+0.59 [0.16, 1.00] rho	5 / 3	AMBIGUOUS
Fig. 6 Lick task and grey screens			
axon response to the square, hit minus miss trials (raw)	+0.12 [0.10, 0.14] dF/F	4 / 1	AMBIGUOUS
axon response to the square, hit minus miss trials (movement removed)	+0.017 [0.002, 0.031] dF/F	4 / 1	AMBIGUOUS
axon fluctuation SD in later subsessions of the lick task (% of subsession 1)	116 [79, 183] %	4 / 1	NOT SEEN
grey screen change across a loom block: axons mean (SD of metric)	+0.31 [-0.50, 1.13] SD	4 / 3	NOT SEEN
grey screen change across a loom block: axons sd (SD of metric)	+0.54 [-0.10, 1.14] SD	4 / 3	NOT SEEN
grey screen change across a loom block: axons rate (SD of metric)	+0.090 [-0.645, 0.825] SD	4 / 3	NOT SEEN
grey screen change (loom minus plain grey): axons mean (SD of metric)	-0.73 [-2.19, 0.08] SD	3 / 2	NOT SEEN
grey screen change (loom minus plain grey): axons sd (SD of metric)	+0.26 [-0.83, 0.94] SD	3 / 2	NOT SEEN
grey screen change (loom minus plain grey): axons rate (SD of metric)	-0.44 [-0.76, 0.00] SD	3 / 2	NOT SEEN
Fig. 7 Controls and limits of the evidence			
across sessions: peak r(axon, face) vs behavioural range (SD of log face motion), Spearman	+0.70 rho	20 / 3	AMBIGUOUS
within sessions: 3 min window coupling vs window behavioural range (Spearman, unflagged windows)	+0.74 rho	155 / 3	AMBIGUOUS
held-out R2 of behaviour model: signal-like axons minus random rejected ROI sets of equal size	+0.13 [-0.04, 0.24] R2	13 / 3	AMBIGUOUS
peak r of the axon mean with the rigid brain shift (registration offsets, lateral motion)	+0.094 [-0.156, 0.333] r	13 / 3	AMBIGUOUS
Data limits (no statistics)			
animals with analysable axon sessions	n/a	20 / 3	NOT AVAILABLE
locomotion (ball) usable	n/a	5 / 1	NOT AVAILABLE
reference channel (tdTomato) / axial-motion control	n/a	-	NOT AVAILABLE
recorded sessions not aggregated (RB22-24)	n/a	19 / 3	NOT AVAILABLE

Verdict rule: SUPPORTED = 95% CI (hierarchical bootstrap over animals / sessions) excludes 0 and most sessions agree; any result carried by one animal is downgraded to AMBIGUOUS automatically; NOT SEEN = CI includes 0 and no session-level support. Effects are in different units and are not comparable with each other. "Unit" columns: r = correlation, z = session z, dF/F = high-passed, SD = SD of the metric over the session's windows. n = 3 mice at most: every number describes these animals.